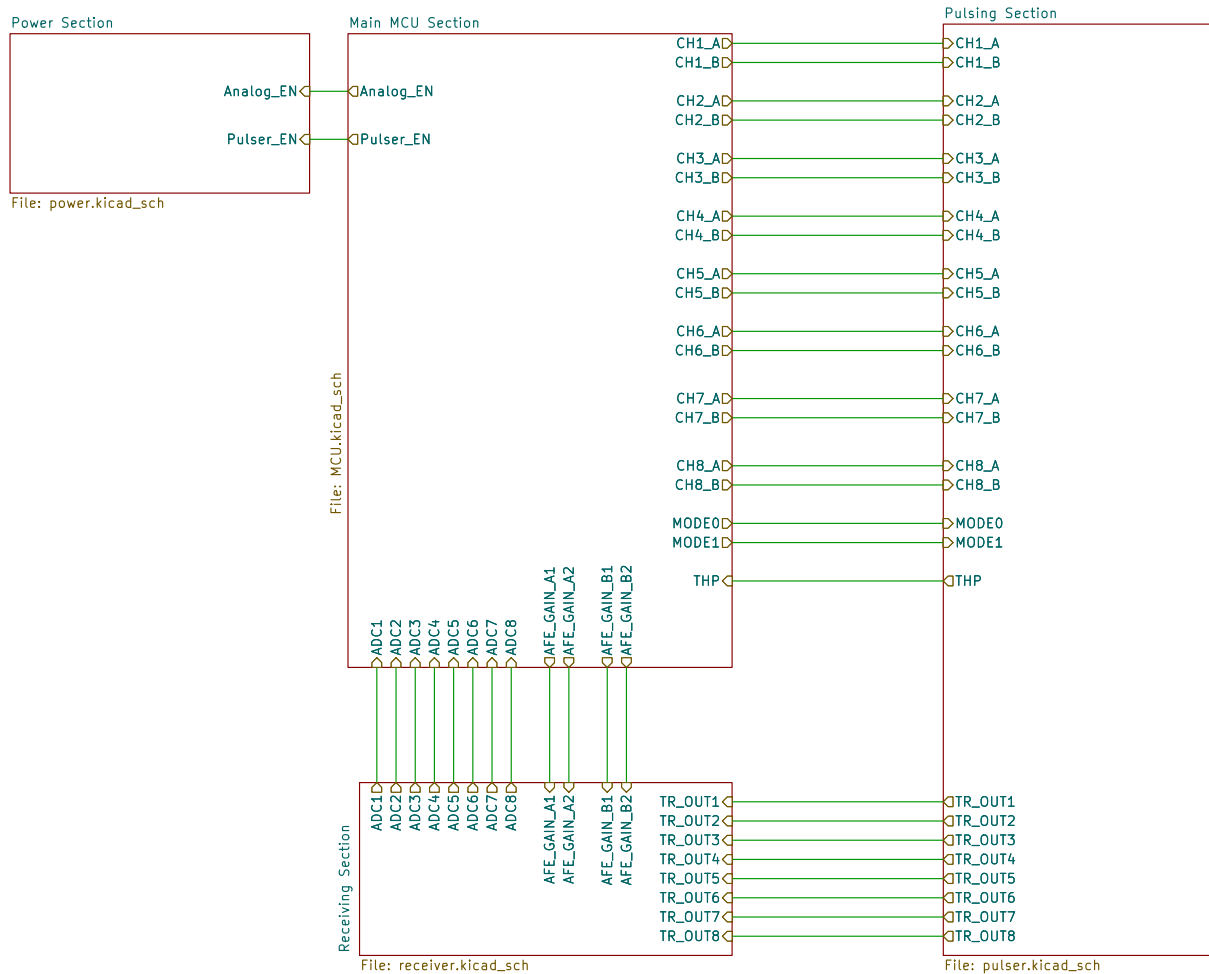
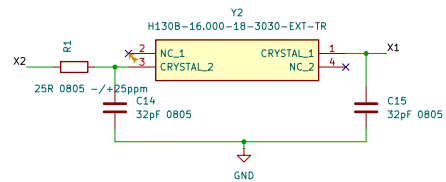




12 Volt (External) (555mA)	7 Volt (Buck Converter) (350mA)	5 Volt (LDO Regulator) (200mA)	LNA + VGA Supply (160mA – short circuit)
			Op-Amps (4 * 10mA = 40mA)
	5 Volt (Buck Converter) (205mA)	3.3 Volt (LDO Regulator) (150mA)	MCU Supply (150mA)
		Positive Pulse Driver Supply (100mA)	
		3.3 Volt (LDO Regulator) (5mA)	Pulser Logic Supply (5mA)
		-5 Volt (Inverting Buck Converter) (100mA)	Negative Pulse Driver Supply (100mA)



$C1 = C2 = 2(C_{load} - C_{stray}) = 2(18 - 2) = 32pF$
 The crystal is required (@ 16MHz):
 - To have a drive level of $\geq 1mW$ (a damping resistor is used)
 - Have maximum ESR of 75 Ohms (50 is used)
 - $12pF < C1 < 24pF$ (18pF is used)

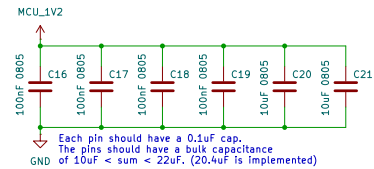
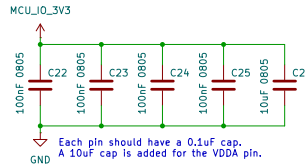
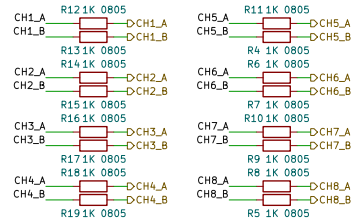
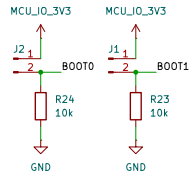


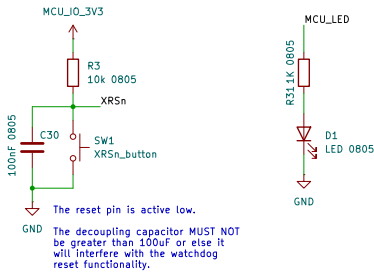
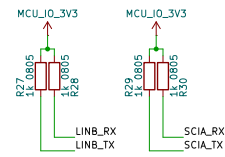
Table 3-1. Device Default Root Modes

BOOT MODE	SPICEN (DEFAULT) BOOT MODE SELECT	SPICEN (DEFAULT) BOOT MODE SELECT	SPICEN (DEFAULT) BOOT MODE SELECT
Factory ID	0	0	0
Factory ID	0	0	0
Factory ID	1	0	0
Factory ID	1	1	1

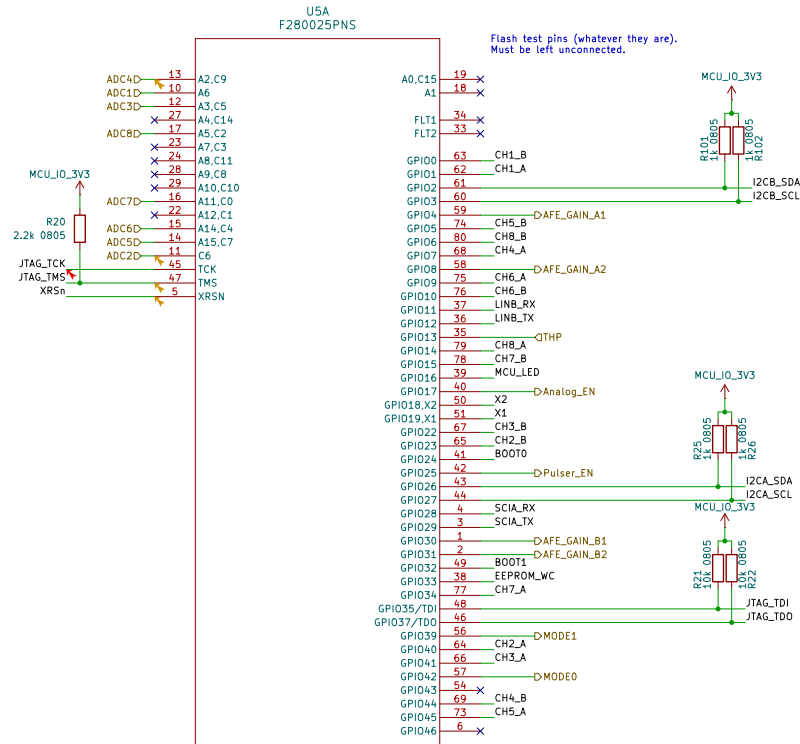
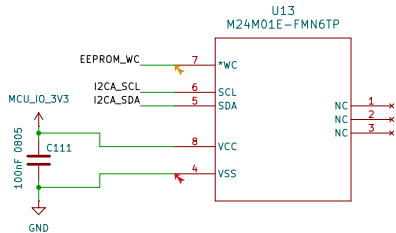


JTAG test mode select (TMS) with internal pullup.
 This serial control input is clocked into the TAP controller on the rising edge of TCK. This device does not have a TMS pin. An external pullup resistor (recommended 2.2 kΩ) on the TMS pin is VDDIO should be placed on the board to keep JTAG in reset during normal operation.

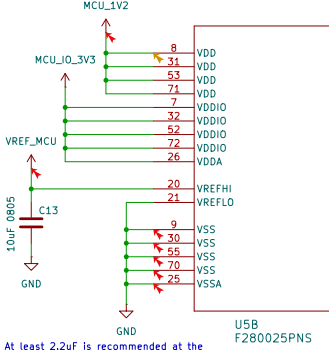
The debugger to be used is the XDS110 debugger (recommended by TI).



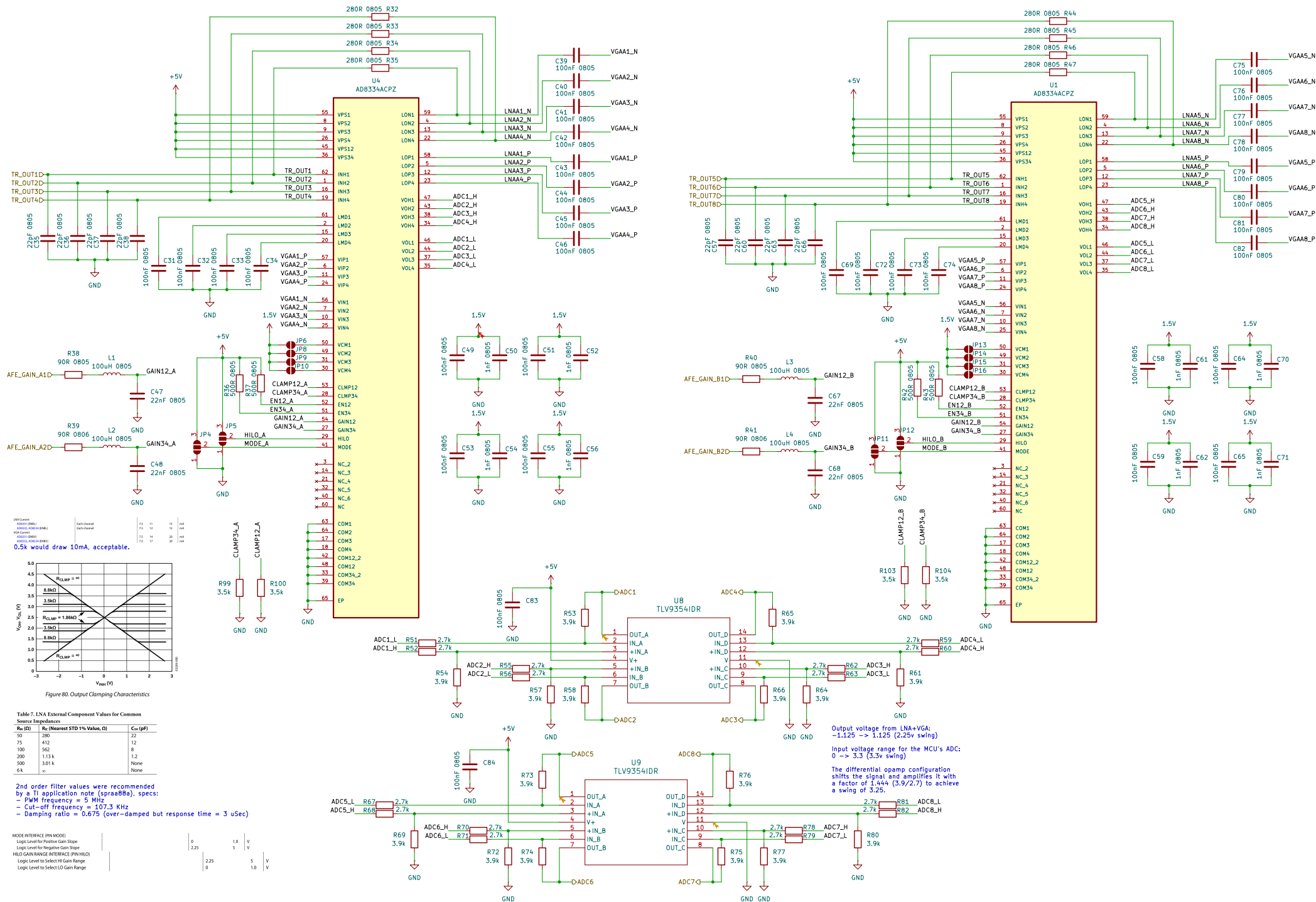
The reset pin is active low.
 The decoupling capacitor MUST NOT be greater than 100uF or else it will interfere with the watchdog reset functionality.



1K resistors are placed at pulser lines to limit current to 3.3mA per pin.



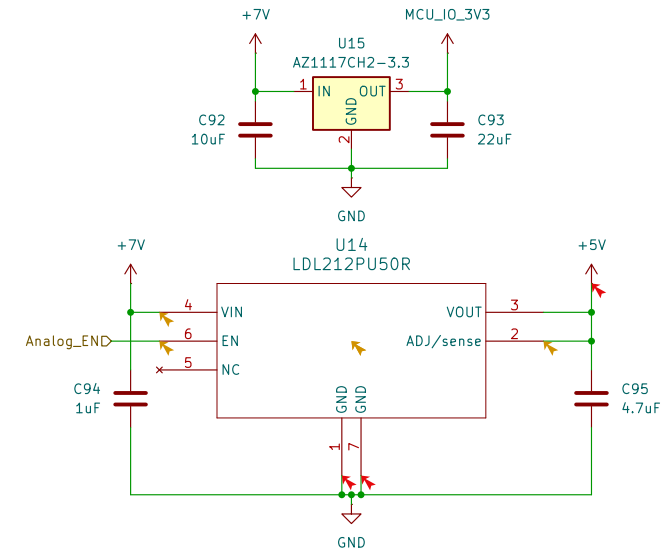
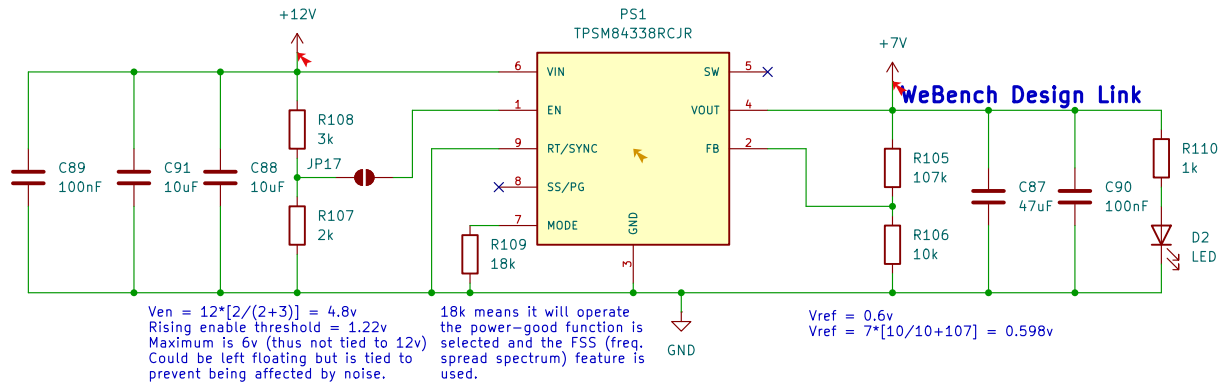
At least 2.2uF is recommended at the Vref_H pin, 10uF is used.



VDD is supplied from an internal 1.2v regulator stepping down from VDDIO but nonetheless its pins need to be decoupled.

In internal VREG mode, tying the VDD pins together is optional as long as each VDD pin has a capacitor connected to pin. See the *VDD Decoupling* section for VDD decoupling configurations.

Add an LDO after the 5v supply to power the ADC from a clean source.
Add an LDO to convert from 5v to 3.3 & 1.2 for the MCU.



Convert the 12v to 5v then use the other converter to convert to -5v.

